KCC CODE CLUB · GAME MAKERS

Week 4 Lesson Plan

Decisions and Sensing

Big idea: The computer can ask a yes-or-no question about the world and do different things depending on the answer.

This is the big week. At the end of today their project stops being an animation and becomes a game they can actually play. Plan for the energy in the room to go up.

At a glance

New concepts	Selection (`if`, `if/else`), booleans as yes/no questions, sensing
New blocks	`if... then`, `if... then... else`, `touching [sprite]?`, `touching [edge]?`, `key [x] pressed?`, `start sound`, `play sound until done`
Class activity	Bumper Sprite
Project checkpoint	Catching works. The game is playable
Homework	Catch feedback (sound or flash); add a bad object

Before the class

- Print the Week 4 handout.
- Skeleton `Sky Catcher — end of Week 3` ready.
- This is the highest-value class in the course — if you're going to have three volunteers in the room on any one week, make it this one.

0:00–0:05 — Show & Tell

Two homework demos. Ask specifically whether anyone got the random falling speed working, and get them to explain *how*, not just show it.

0:05–0:15 — Live demo: if, if/else, and touching

Build live:

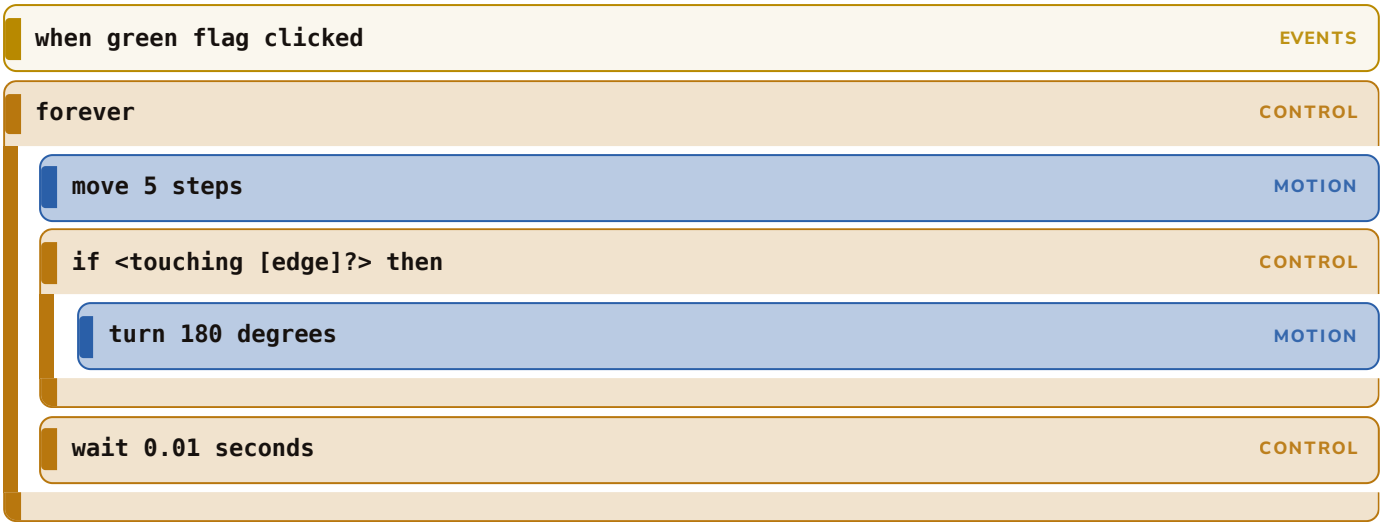
1. `when green flag clicked` → `forever [ if <touching [edge]?> then [ say [ouch] for 1 second ] ]` with the sprite draggable. Drag it into the edge. It complains.
2. Point at the **shape** of the blocks. The `touching?` block is a pointy hexagon; the slot in the `if` block is a pointy hexagon. Shapes tell you what fits where. This is a genuinely useful piece of Scratch literacy and it takes ten seconds.
3. Add an `else`: `if <touching edge?> then [say ouch] else [say fine]`.
4. Swap in `touching [Sprite2]?` and drag two sprites together.

5. **Crucially:** show what happens when the `if` is *not* inside a `forever`. It checks once, at the moment the flag is clicked, finds nothing, and gives up forever. Run it, look confused, then find it with them.

Number 5 is the bug that will consume half your floor time today. Front-load it.

0:15–0:30 — Class activity: Bumper Sprite

It's an `if` inside a `forever`, which is exactly the shape the catching script needs in the project build. That's why this one stays.



Add a second sprite and have them bounce off each other too. Kids will spend the whole fifteen minutes on this happily — it's the first thing that behaves like it has a mind of its own.

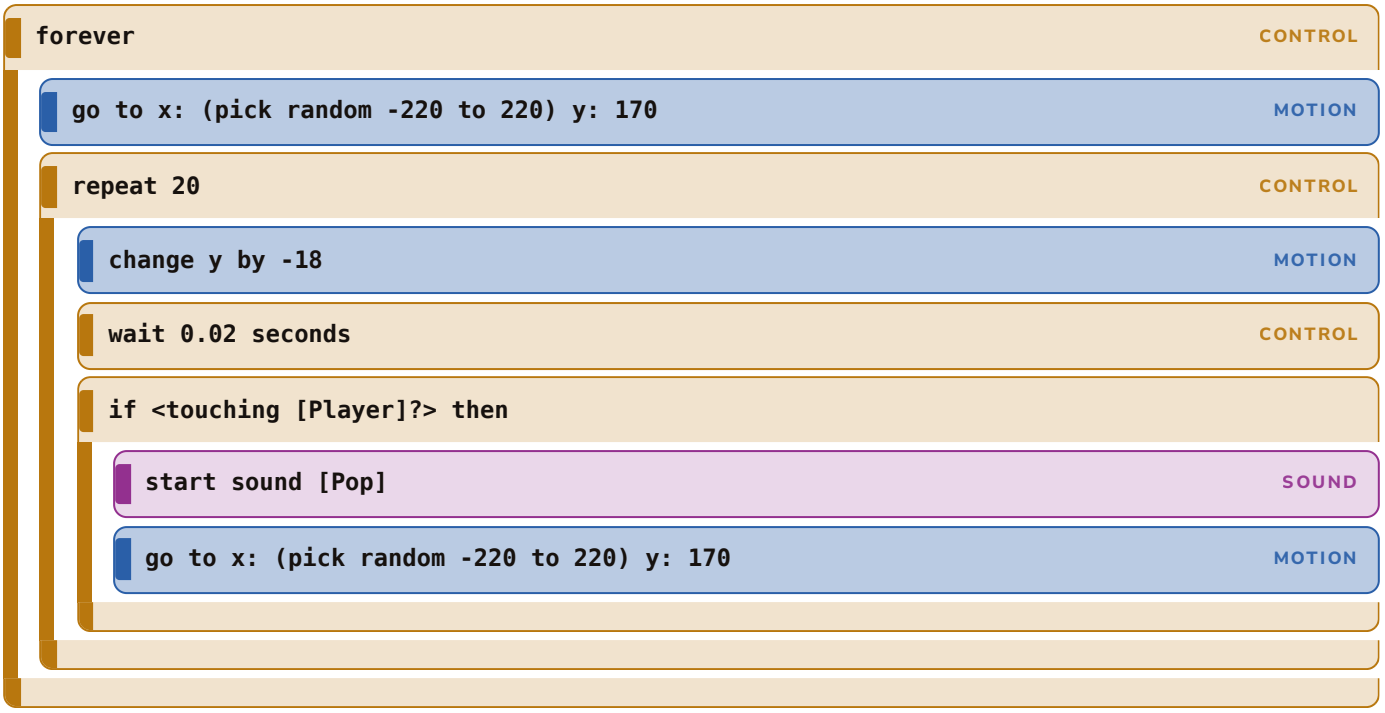
0:30–0:55 — Project build: catching

Goal: **"By the end of today, you can catch things and the game will notice."**

Blocks on the board: `if... then`, `touching [player]?`, `start sound`, and the ones they already have.

The change goes **inside the falling loop**, which is the part they'll get wrong:





The `go to` inside the `if` is what makes the caught object jump back to the top immediately instead of continuing to fall through the player.

Let them struggle with placement for a few minutes before you help. The two questions to ask:

- *"When does your computer check whether you've caught it?"* (Answer: only when the `if` runs.)
- *"How many times does it check as the object falls?"* (Answer: once per trip through the repeat loop — twenty times.)

Then **playtest**. Everyone plays their own game for two minutes and tunes the speed. Then swap seats and play their neighbour's for two minutes. The room will be loud. That's the point.

Save.

0:55–1:00 — Wrap

Save and back up carefully this week; there's more to lose now than there was.

Tell them explicitly: *"You made a game. It has no score yet, so it's a bit pointless — that's next week."*

Homework

1. **Catch feedback.** When you catch something, make it do something noticeable first — a different sound, a costume flash, a `say`, or a size change.
2. **Something you shouldn't catch.** Add a bad falling object (a bomb, a coconut, a cat). For now, when it touches the player it just says "Ouch!" — next week it'll cost you points.

Bonus challenges

1. **Traffic Light.** Add a red sprite. Then `forever` → `if <touching [Red]?> then say [STOP!] else say [GO!]`, and drag your sprite in and out of the red one.
2. Make a sprite that follows the mouse pointer, but only while the space key is held down.

3. Use `if` to stop the player leaving the stage — the thing we couldn't do in week two.
4. Combine two conditions with `and`: only catch if you're touching it **and** the space key is pressed. Now it's a "grab" game.

Common bugs this week

Symptom	Cause
Catching never works	The `if` isn't inside the `forever` / `repeat` loop
Catching works once, then never again	The `if` is in the outer `forever` but outside the falling `repeat`
Object passes through the player	The `if` fires but there's no `go to` to move it away, so it keeps falling
Catching works but is very fussy	The falling step is too big — it jumps past the player between checks. Smaller `change y by`, more repeats
`touching?` is checking the wrong thing	The dropdown defaults to the first sprite alphabetically. Check it
Sound doesn't play	No sound added to that sprite yet — Sounds tab, choose from the library

The "very fussy" one is subtle and worth understanding: if the object moves 18 pixels between checks, there are gaps where it's past the player without ever having touched it. Halving the step and doubling the repeats fixes it, and it's a lovely accidental introduction to the idea that computers only see the world at the moments they look.

Notes for the second volunteer

Expect the room to fragment today — some will have catching working in five minutes and some will still be placing the `if` at the twenty-minute mark. Get the fast ones to help their neighbours rather than racing ahead into bonus challenges. Today of all weeks, nobody should go home without a working catch.

Links

- Don't fall in! — Raspberry Pi Foundation, a natural next game
- Skeleton: `Sky Catcher — end of Week 4` in the club studio